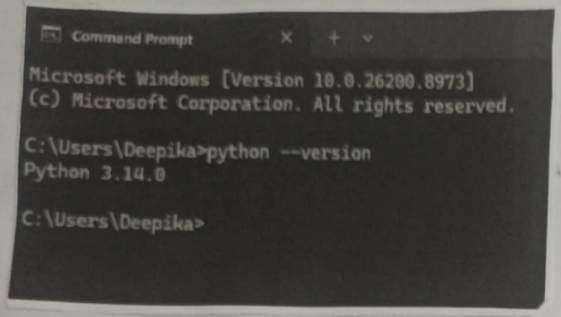
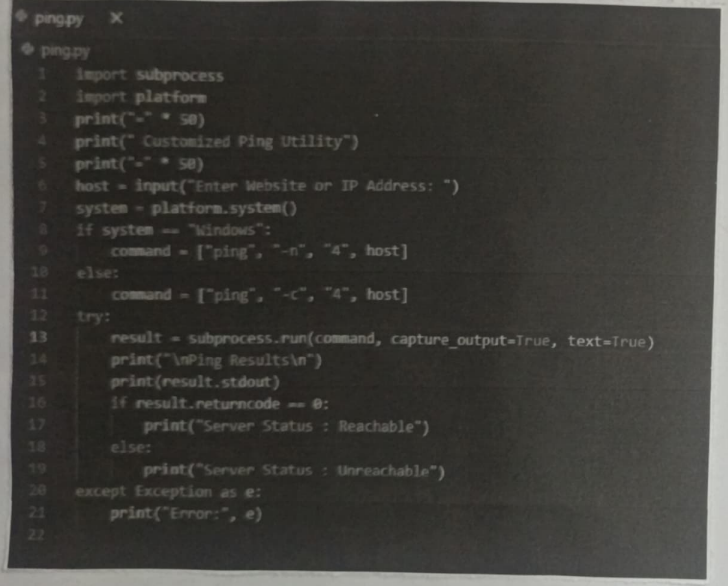
EX NO:6 DEVELOP A CUSTOMIZED PING COMMAND TO
DATE: 01/08/26 TEST THE SERVER CONNECTIVITY





AIM:

To develop a customized ping command in python that checks the connectivity between a client & a server by sending ICMP Echo request packets and displaying the response time and status.

THEORY:

A ping command is a network diagnostic tool used to determine whether a host (server) is reachable over a network. It works using the ICMP (Internet control Message protocol).

The customized ping program allows the user to:

- Enter any hostname
- Send multiple ping requests
- Show packet loss.

PROCEDURE:

1. Install python & VS code
2. Create a python file named ping.py
3. Import the requirement modules
4. Accept the hostname or Ip address from the user.
5. Detect the OS

```
PS C:\Users\Deepika\Downloads\CustomizedPing> python ping.py
==========================================
 Customized Ping Utility
==========================================
Enter Website or IP Address: google.com

Ping Results


Pinging google.com [142.251.221.174] with 32 bytes of data:
Reply from 142.251.221.174: bytes=32 time=5ms TTL=119
Reply from 142.251.221.174: bytes=32 time=5ms TTL=119
Reply from 142.251.221.174: bytes=32 time=7ms TTL=119
Reply from 142.251.221.174: bytes=32 time=6ms TTL=119

Ping statistics for 142.251.221.174:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 7ms, Average = 5ms

Server Status : Reachable
```

```
PS C:\Users\Deepika\Downloads\CustomizedPing> python ping.py
==========================================
 Customized Ping Utility
==========================================
Enter Website or IP Address: ajfbjegeufeugwegeg.com

Ping Results

Ping request could not find host ajfbjegeufeugwegeg.com. Please check the name and try again.

Server Status : Unreachable
```

6. Use ping -n 4 on Windows or ping -c 4 on Linux / macOS
7. Execute the ping command using the subprocess module
8. Display the ping o/p, including reply time & packet statistics
9. Check the return code to determine whether the server is reachable
10. Display Reachable (or) Unreachable based on the result

o/p verified

RESULT:

Thus the customized ping command to test the server connectivity is successful executed and verified.